Lab 3: Architectural Pattern Selection and Component Modelling

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Architecture Selection

We chose Layered Architecture for the Customizable Subscription Box Scheduler.

Reason 1

The system has clear responsibilities: the Subscriber Portal handles user interactions, the business components manage customization, pause/skip rules and fulfillment generation, and the data layer stores subscriber, subscription and box information. This separation makes the system easier to maintain and test.

Reason 2

The architecture suits the application's centralized subscription workflow. The 48-hour customization rule, preference-tag filtering, recurring billing, and fulfillment manifest generation can be coordinated through well-defined business services without the operational complexity of deploying many independent microservices.

Security Advantage

The Payment Gateway is isolated behind a dedicated payment interface. Subscriber personal and payment-related data can be restricted to authorized services, while TLS 1.2+ protects communication and AES-256 protects stored data, matching the security requirement from Lab 1.

Performance Benefit

Frequently accessed menu, preference and subscription data can be retrieved from the data layer with minimal overhead. The Fulfillment Manifest component can generate shipping labels in a dedicated process, supporting the requirement to export labels for 10,000 monthly boxes in under one minute.

Components Identified

Subscriber Portal, Subscription Manager, Box Customization, Fulfillment Manifest, Subscription Database, and Payment Gateway.

Interfaces Shown

Subscription API, Box Service API, Manifest API, Database Queries, Save Selections, Read Confirmed Boxes, and Billing API (shown as UML ball-and-socket assembly connectors in draw.io; see component_diagram.drawio).